Final Project Phase I Presentation

Team: The Stakeholders

Members: Aryan Puthran, Vijay Deva, and Julia Smith

Scenario 1

AS-IS

- Stevie tries to walk to his next classroom.
- He knows he must walk forward a few steps and turn left around the corner.
- Stevie is unsure of the exact turning point, guesses incorrectly, turns too early, and strikes the wall with his head.

TO-BE

- The THEIA app asks Stevie to provide his current location and destination.
- The app calculates the indoor path. THEIA delivers real time voice cues instructing Stevie to walk forward a set distance and prompts the exact moment to turn left.

Scenario 2

AS-IS

- A blind student enters an unfamiliar campus hall and needs to find Room 214.
- The student taps along the wall with a white cane.
- Door frames lack accessible braille signs at reaching height.
- The student stops and waits for a passerby to ask for verbal directions that end up confusing.

TO-BE

- The blind student opens THEIA and speaks the command "Navigate to Room 214".
- THEIA confirms the room, calculates the path from the building entrance, loads the floor layout, and voices step-by-step route guidance through an earpiece.

Scenario 3

AS-IS

- Marcus has low vision and enters an administrative building to reach financial aid.
- The hallway lighting is dim and door plaques are illegible.
- Marcus tries using his smartphone screen, but the small default interface causes severe eye strain and disorients him.

TO-BE

- Marcus opens THEIA to locate the office.
- The app displays an interface with high contrast bold visuals and magnified text.
- Marcus taps the large high contrast tactile prompt to select administrative offices, receiving immediate audio confirmations alongside clear directional cues.

Scenario 4

AS-IS

- Elena walks near an indoor stairwell during a busy class change.
- Elena slips on a slick surface, falls, and loses her physical orientation.
- Elena cannot reach her phone inside her backpack to request help and remains stranded on the floor waiting for assistance.

TO-BE

- Elena slips and falls near the stairwell.
- The smartphone accelerometer registers the impact.
- THEIA emits an audible prompt asking Elena to confirm her status.
- When Elena does not respond within thirty seconds, the app sends an emergency distress signal with her indoor coordinates to campus security.

Scenarios & Features

- Precise Indoor Turn Navigation
  - Features: Real time step distance cues and accurate turn timing.
  - Benefitting Group: Visually impaired students walking through complex hallway corners.
- Initial Room Wayfinding
  - Features: Voice command input and floor layout loading.
  - Benefitting Group: Blind visitors and students going through unfamiliar buildings.
- Low Vision Visual Accessibility
  - Features: High contrast interface and large button inputs.
  - Benefitting Group: People who have some vision.
- Automated Emergency Response
  - Features: Fall detection and automated distress messaging and communication.
  - Benefitting Group: Users who are visually impaired that are lost or injured, and by themselves.

Priority TO-BE Scenario & Analysis I

- Priority Selection: Precise Indoor Turn Navigation
  - This is an important scenario because visually impaired users use directional turns constantly during any indoor trip.
  - Not guiding a user through a turn leads to collisions and even injuries, making turn accuracy the most critical task of the system.
- Who initiates and interacts?
  - The initiator is the primary user that requests the route to the classroom.
  - The THEIA navigation engine acts as a guide, continuously tracking position and prompting cues without requiring the user to have to tap the screen constantly during movement.

Priority TO-BE Scenario & Analysis II

- What kinds of turns and environments do we support?
  - Supported turns include 90 degree hallway corners, hallway intersections, and even entryway turns.
- When are turns prompts given to the user?
  - THEIA voices a warning in advance, 2 meters before a corner, which is then followed by a prompt at the time of the turn.
  - If the user continues walking straight past the corner, the system detects this and warns the user to stop.

Priority TO-BE Scenario & Analysis III

- How to resolve movement conflicts?
  - If another person or object blocks the corner, the obstacle detection sensor alerts the user to either wait or take a sidestep before they turn.
  - If a smartphone walking sensor miscounts strides, the app will recalibrate indoor coordinates using Wi-Fi if available, or Bluetooth if not.
- What-if scenarios.
  - What if the user walks faster/slower than average? THEIA updates turns based on real time walking speed instead of a set amount.
  - What if cues cannot be heard due to ambient noise? The phone vibrates a pattern to notify the user to turn.
  - What if the smartphone loses tracking due to connection loss or for some other reason? The system tells the user to stop, and attempts to get a signal back.